

Sensitivity of optimal replacement policies to lifetime parameter estimates

Mitra Fouladirad^{a,*}, Christian Paroissin^b, Antoine Grall^a

^a*Laboratoire de Modélisation et Sécurité des Systèmes
Université de Technologie de Troyes, Troyes, France*

^b*CNRS / Univ Pau & Pays Adour, Laboratoire de Mathématiques et de leurs applications
de Pau fédération IPRA, UMR 5142, 64000 Pau, France.*

Abstract

This paper proposes to analyze three time-based replacement policies when the parameters of the time-to-failure distribution are unknown. Under the hypothesis of a large sample data, the unknown parameters are estimated via maximum likelihood method. The uncertainty around the point estimates impacts the maintenance decision rules. Based on the asymptotic normality of the parameter estimators and delta-method, the asymptotic distribution of the optimal decision parameter and the average cost is derived. This distribution permits to measure the uncertainty around the optimal maintenance decision rule induced by the point estimates. The sensitivity analyze of the replacement policies to the maintenance unit costs and lifetime distribution parameters, is carried out. Application to two time-to-failure distributions is provided.

Keywords: Maintenance, Age replacement policy, periodic replacement policy, block replacement policy, Delta-method, Lifetime Parameters estimation

1. Introduction

Maintenance optimization is an important issue in reliability and industrial engineering. In order to avoid failure and its undesirable consequences, preventive maintenance actions are planned. Time-based preventive maintenance is the first type of preventive maintenance policy proposed in the literature and used in practice [10]. Without any information on the systems health and age, preventive actions are often planned periodically, refer to [10, 17, 19]. In presence of information on the system, maintenance strategies plan preventive actions based either on system deterioration level or on the system age (effective

*Corresponding author. Postal address: Université de Technologie de Troyes, 12 rue Marie Curie, CS 42060, 10004 Troyes CEDEX, France. Tel: +33 3 25 71 80 72

Email addresses: `mitra.fouladirad@utt.fr` (Mitra Fouladirad), `cparoiss@univ-pau.fr` (Christian Paroissin), `antoine.grall@utt.fr` (Antoine Grall)

or virtual age), refer to [10, 14, 22]. One of the main objectives of maintenance optimization is to propose maintenance preventive dates which permit to avert failure while avoiding to frequent preventive actions. The optimization criteria is often a long run average cost or availability evaluated from system lifetime distribution and maintenance unit costs.

For a given parametric distribution of time-to-failure (or lifetime) of a device, one can determine (at least numerically) the optimal replacement period for a replacement policy under consideration. Therefore, this optimal replacement period depends on the parameters of the time-to-failure distribution. Most of the works on time-based preventive maintenance optimization such as [10, 21] consider that the lifetime distribution parameters are known. Nevertheless, in practice, these parameters are often unknown. In presence of a sample time-to-failure observations, the parameters can be estimated. The estimation procedure and its precision mainly depend on the size of the available sample. When a large sample is available, classical statistical inference methods such as maximum likelihood methods can be used whereas in presence of a small sample size enriched with expert judgement Bayesian methods are recommended [7, 23]. In absence of expert judgement, Bayesian method can also be applied by considering a reference or non-informative prior [24, 16]. In any case, the optimal maintenance policy or more precisely the optimal replacement period depends on parameters estimate. Consequently, the obtained optimal replacement period is again an estimate of the real optimal value. A question that arises naturally is how to measure the variability of the optimal replacement period induced by the estimation procedure. More precisely, how to evaluate the bias between the obtained optimal replacement period or cost and the real optimal values. Some papers tried to respond to this question for instance in presence of a small sample authors in [20, 26, 27] use a Bayesian framework to deal with the parameter uncertainty. The results give uncertainty around the decision parameter but no thorough sensitivity analysis is carried out. In de Jonge et al. [11, 12] authors treat the problem by associating a distribution to unknown parameters in a time based maintenance policy framework. The results highlight the impact of the uncertainty on the optimal results. To propose such a distribution on parameter, prior knowledge on this latter is necessary. In absence of such information it is difficult to propose a suitable distribution. In any case, the results point out that the uncertainty cannot be ignored. In the same context, when samples with different sizes are available, Baker and Scarf [2] treat the problem of uncertainty on maintenance decision variables by studying the impact of the sample size. They show that the excess cost due to parameter estimators bias is inversely proportional to data sample size. Which means that even if parameter estimates are imprecise, the excess cost is not so great. But de Jonge et al. [11] showed that sometimes ignoring the uncertainty can be very costly even for small levels of uncertainty. Other works, such as Bedford et al. in [4], tackle the problem of the uncertainty around model parameters in decision making and failure prediction. Bedford and Buena [3] studied the impact of censored reliability data due to age-based maintenance on decision variables estimation but they don't discuss the impact of the parameter uncertainty on the decision

procedure. In sum, all these papers emphasize that consideration of uncertainty in maintenance optimization is crucial and this fact opens avenue to new studies in this field.

In this paper, one considers a large size data set is available. This case can refer to data available on items or systems lifetime during decades. For instance historical electricity operators or railways companies have in their disposal a large sample of lifetime data for their components from the early age of establishment until present. In this framework the maximum likelihood estimator and its asymptotic properties can be used to propose confidence intervals around the maintenance parameter estimation for three replacement policies to the uncertainty. In contrary to Baker and Scarf in [2] the aim is not to analyze the impact of the sample size on the replacement policies, but under the hypothesis of a very large sample size the aim is to give asymptotic distribution of the decision parameters. One proposes exact asymptotic expressions of the dispersion of the optimal cost and decision variable with respect to the lifetime distribution function. In other words, for each policy, the standard deviation of the bias associated to the estimated optimal cost and period are given. Thus the risk aversion of each policy can be measured and their confidence degree can be evaluated.

In Section 2, first some well-known results for the replacement policies are recalled. Then, some general results on the sensitivity analysis of these replacement policies are provided. These results are based on the assumption of asymptotic normality of the parameter estimators. It allows to apply the classical delta-method and some of its extensions. In Section 3, applications to two time-to-failure distributions are provided.

2. Replacement policies: preliminaries

In this paper, one shall consider three different replacement policies, namely the periodic replacement policy, the age replacement policy and block replacement policy (refer to [21] or [28]). One shall denote T the time-to-failure (or lifetime) of the device. It is assumed that the distribution of T depends on some parameter $\theta \in \Theta \subset \mathbb{R}^p$. One considers the usual notation: $f_T(\cdot; \theta)$ for its probability density function, $F_T(\cdot; \theta)$ for its cumulative distribution function, $S_T(\cdot; \theta)$ for its survival function (or reliability), $h_T(\cdot; \theta)$ for its hazard rate (or failure rate) and $M_T(\cdot; \theta)$ for its renewal function.

The three maintenance policies under consideration depend on a single decision variable τ , the duration between two consecutive inspections or replacements which induces a certain cost. Thus, one can be interested in determining the optimal replacement period τ^* . In order to determine it, one considers the asymptotic cost per unit of time defined as follows:

$$C(\tau; \theta) = \lim_{t \rightarrow \infty} \frac{C(\tau, t; \theta)}{t},$$

where $C(\tau, t; \theta)$ is the cost over the time interval $[0, t]$ when the device is inspected and perhaps replaced at $(k\tau)_{k \in \mathbb{N}}$. As it is well known (see [13], for

instance), according to the renewal theory, one has:

$$C(\tau; \theta) = \frac{\text{Expected cost over a cycle}}{\text{Expected cycle length}}.$$

2.1. Periodic replacement policy

In this paper, a periodic replacement policy is considered where the device is replaced by a new one periodically and there is no replacement at time-to-failure. There exist two different costs : the cost c_r for replacing the device by a new one and the unavailability time unit cost c_u induced during the period after failure and before the replacement. For such policy,

$$C(\tau; \theta) = \frac{\mathbb{E}[c_r + c_u(\tau - T)_+]}{\tau} = \frac{c_r + c_u \int_0^\tau F_T(u; \theta) du}{\tau}.$$

Moreover, $\lim_{\tau \rightarrow 0} C(\tau; \theta) = +\infty$ and $\lim_{\tau \rightarrow +\infty} C(\tau; \theta) = c_u$.

Let us denote $\tau^* := \operatorname{argmin}_{\tau > 0} C(\tau; \theta)$. Differentiating the above expression of the cost function $C(\tau; \theta)$, τ^* is the root of the following function (with respect to τ):

$$\phi(\tau; \theta) = \int_0^\tau u f_T(u; \theta) du - \frac{c_r}{c_u} = -\tau S_T(\tau; \theta) + \int_0^\tau S_T(u; \theta) du - \frac{c_r}{c_u} \quad (1)$$

(recalling that $\frac{c_r}{c_u} \leq 1$). The solution depends on the two costs c_r and c_u only through their ratio (and so does not depend on the monetary unit). Since, for fixed $\theta \in \Theta$, $\tau \mapsto \phi(\tau; \theta)$ is an increasing function towards $\mathbb{E}[T] - c_r/c_u$ (which depends on θ) with $\phi(0; \theta) = -c_r/c_u$, it follows that τ^* is finite if $\mathbb{E}[T] > c_r/c_u$ and is infinite otherwise. Under this condition of existence for τ^* , and provided that $t \mapsto F_T(t; \theta)$ is a one-to-one application, it is well-known (see, e.g., [21], page 120) that the optimal cost satisfies the following relation :

$$C^* = C(\tau^*; \theta) = c_u F_T(\tau^*; \theta) \Leftrightarrow \tau^* = F_T^{-1}(C^*/c_u; \theta), \quad (2)$$

where $F_T^{-1}(\cdot; \theta)$ is the quantile function of the random variable T . Since $\phi(\tau^*; \theta) = 0$ and replacing τ^* by its expression given in Equation (2), C^* an implicit function ψ is obtained where $\psi(C^*; \theta) = 0$:

$$\int_0^{F_T^{-1}(C^*/c_u; \theta)} u f_T(u; \theta) du - \frac{c_r}{c_u} = 0 \Leftrightarrow \psi(C^*; \theta) = \int_0^{C^*/c_u} F_T^{-1}(v; \theta) dv - \frac{c_r}{c_u} = 0 \quad (3)$$

2.2. Age replacement policy

Here it is assumed that the device is replaced either at failure or at age τ if no failure occurs between 0 and τ ($\tau = \infty$ means that replacement occurs only at failure). Such scheme is so-called age replacement policy. In this framework , there are two different costs: c_f the one incurred by the replacement of a failed

device and $c_{nf} < c_f$ for the preventive replacement. For such policy, the average cost is as follows:

$$C(\tau; \theta) = \frac{c_f F_T(\tau; \theta) + c_{nf} S_T(\tau; \theta)}{\int_0^\tau S_T(u; \theta) du}.$$

When τ tends to infinity, and since such case corresponds to replacements only at failure, then it can easily be seen that $C(\tau; \theta)$ tends to $C(\infty; \theta) := c_f / \mathbb{E}[T]$.

Let us denote $\tau^* := \operatorname{argmin}_{\tau > 0} C(\tau; \theta)$. According to Theorem 3.2 in [21], if T has a strictly increasing and continuous failure rate and if $h_T(\infty; \theta) > \frac{1}{\mathbb{E}[T]} \frac{c_f}{c_f - c_{nf}}$, then τ^* is the unique and finite root of the following function (with respect to τ):

$$\phi(\tau; \theta) := h_T(\tau; \theta) \int_0^\tau S_T(u; \theta) du + S_T(\tau; \theta) - \frac{c_f}{c_f - c_{nf}}. \quad (4)$$

Nakagawa ([21], page 74) has proved that, for fixed $\theta \in \Theta$, $\tau \mapsto \phi(\tau; \theta)$ function is strictly increasing when $h_T(\cdot; \theta)$ is continuous and strictly increasing (this assumption is quite reasonable since such policy is relevant only for ageing devices). Moreover, under this assumption, the optimal cost and the optimal age are related as follows:

$$C^* = C(\tau^*; \theta) = (c_f - c_{nf}) h_T(\tau^*; \theta) \Leftrightarrow \tau^* = h_T^{-1}(C^* / (c_f - c_{nf}); \theta) \quad (5)$$

Replacing this expression of τ^* in Equation (4), it leads to an implicit expression of C^* in function of parameter θ :

$$\begin{aligned} \psi(C^*; \theta) &:= \frac{C^*}{c_f - c_{nf}} \int_0^{h_T^{-1}(\frac{C^*}{c_f - c_{nf}})} S_T(u; \theta) du \\ &+ S_T(h_T^{-1}(C^* / (c_f - c_{nf}); \theta); \theta) - \frac{c_f}{c_f - c_{nf}} = 0. \end{aligned} \quad (6)$$

2.3. Block replacement policy

Here it is assumed that the device is replaced either at failure or periodically at τk with $k \in \{1, 2, \dots\}$. The replacement is supposed to be instantaneous. Such scheme is so-called block replacement policy and is more often used for multi-component systems. In this framework, there are two different costs: c_f the one incurred by the replacement of a failed device and $c_{nf} < c_f$ for the preventive replacement. For such policy, the average cost during a period τ is as follows:

$$C(\tau; \theta) = \frac{c_f \mathbb{E}(N(\tau)) + c_{nf}}{\tau},$$

where $N(t)$ is the number of failures occurred in interval $[0, t]$. Let us denote $M(t; \theta) = \mathbb{E}(N(t))$ the renewal function. If the device is replaced only at failure, it means τ tends to infinity, then it can be easily seen that $C(\tau; \theta)$ tends to $C(\infty; \theta) := c_f / \mathbb{E}[T]$. To find an optimum planned replacement period

τ that minimizes $C(\tau; \theta)$. It is assumed that $M(t; \theta)$ is differentiable and let be $m(t; \theta) = \partial M(t; \theta) / \partial t$, where $m(t; \theta)$ is called the renewal density. Then, differentiating $C(\tau; \theta)$ with respect to τ and setting it equal to zero, we have

$$\phi(\tau; \theta) = \tau m(\tau; \theta) - M(\tau; \theta) - \frac{c_{nf}}{c_f} \quad (7)$$

and therefore

$$C^* = C(\tau^*; \theta) = c_f m(\tau^*; \theta) \Leftrightarrow \tau^* = m^{-1}(C^* / c_f; \theta).$$

Replacing this expression of τ^* in Equation (7), it leads to an implicit expression of C^* in function of parameter θ :

$$\psi(\tau^*; \theta) = C^* m^{-1} \left(\frac{C^*}{c_f}; \theta \right) - M \left(m^{-1} \left(\frac{C^*}{c_f}; \theta \right); \theta \right) - c_{nf} = 0 \quad (8)$$

3. General results for the sensitivity analysis

In this section, first the general framework is established and essential tools (theorems) for the sensitivity analysis to parameter estimates uncertainty are introduced. Afterward, the impact of parameter estimates uncertainty on the two replacement policies is carried out. For each policy the use of the proposed tools is discussed.

3.1. General framework

Let us denote by $\theta_0 \in \Theta$ the true parameter of the time-to-failure distribution. As seen above, the optimal replacement period τ_0^* , can be computed by determining the root of $\phi(\cdot; \theta_0)$. Most of the time, the parameter θ_0 is unknown but could be estimated from a sample of time-to-failure observations. Let us denote by $\hat{\theta}_n$ an estimator of θ , say for instance the maximum likelihood estimator. Assume that this estimator has 'good' properties, like consistency:

$$\hat{\theta}_n \xrightarrow[n \rightarrow \infty]{Pr} \theta_0, \quad (9)$$

and like asymptotic normality:

$$\sqrt{n} (\hat{\theta}_n - \theta_0) \xrightarrow[n \rightarrow \infty]{d} \mathcal{N}(0, \Sigma_0^2), \quad (10)$$

where the asymptotic variance-covariance matrix Σ_0^2 depends on θ_0 . If θ_0 is unknown, one can replace θ_0 by $\hat{\theta}_n$ in order to estimate τ_0^* . In such case, there is only an estimator $\hat{\tau}_n^*$ of the optimal replacement period and an estimation of total cost denoted by $\hat{C}_n^*$.

In this framework, we aim to study the asymptotic properties of $\hat{\tau}_n^*$, $\hat{C}_n^*$ and the bias $|\hat{C}_n^* - C_0^*|$.

First consider the estimated cost function $C(\cdot; \hat{\theta}_n)$ obtained by the plug-in method. By a simple application of the continuous mapping theorem and provided that Equation (9) is satisfied, $C(\tau; \hat{\theta}_n)$ is a convergent (point-wise) estimator of $C(\tau; \theta_0)$:

$$\forall \tau > 0, \quad C(\tau; \hat{\theta}_n) \xrightarrow[n \rightarrow \infty]{Pr} C(\tau; \theta_0).$$

A convenient tool to prove the asymptotic normality of an estimator is the so-called delta-method, refer to [25, 9]. It can be stated as follows:

Theorem 1. *Let $(X_n)_{n \in \mathbb{N}^*}$ be a sequence of $\mathbb{R}^p$ -valued random vectors. Assume there exists $\mu_X \in \mathbb{R}^p$ and Σ a definite positive matrix such that*

$$\sqrt{n}(X_n - \mu_X) \xrightarrow[n \rightarrow \infty]{d} N(0, \Sigma).$$

Let q real functions $f_1, \dots, f_q$ with continuous first partial derivatives at μ_X , where at least one of these derivatives is non-zero. For $i \in \{1, \dots, q\}$ and for any $n \in \mathbb{N}^$, set $Y_{i,n} = f_i(X_n)$, $Y_n = (Y_{1,n}, \dots, Y_{q,n})^T$ and $\mu_Y = (f_1(\mu_X), \dots, f_q(\mu_X))^T$. Then, the sequence $(Y_n)_{n \in \mathbb{N}^*}$ is also asymptotically normal:*

$$\sqrt{n}(Y_n - \mu_Y) \xrightarrow[n \rightarrow \infty]{d} N(0, K\Sigma K^T),$$

where K is the $q \times p$ matrix with elements $k_{i,j} = \partial f_i / \partial x_j$ for $i \in \{1, \dots, q\}$ and $j \in \{1, \dots, p\}$.

Next theorem states that the asymptotic normality of $C(\tau; \hat{\theta}_n)$ holds by application of the classical delta-method (a direct application of delta method).

Theorem 2. *Assume that Equation (10) is satisfied and that $\theta \mapsto C(\tau; \theta)$ is differentiable for any $\tau > 0$. Let $\nabla_\theta C(\tau; \theta)$ be the gradient vector of the cost function (with respect to θ). If $\nabla_\theta C(\tau; \theta)$ is continuous and if $\nabla_\theta C(\tau; \theta_0) \neq 0_{\mathbb{R}^p}$, then $C(\tau; \hat{\theta}_n)$ is an asymptotic normal (point-wise) estimator of $C(\tau; \theta_0)$:*

$$\forall \tau > 0, \quad \sqrt{n} \left(C(\tau; \hat{\theta}_n) - C(\tau; \theta_0) \right) \xrightarrow[n \rightarrow \infty]{d} \mathcal{N}(0, \sigma_{cost}^2),$$

with $\sigma_{cost}^2 = \nabla_\theta C(\tau; \theta_0) \Sigma(\theta_0) \nabla_\theta C(\tau; \theta_0)^T$.

Unfortunately, one cannot apply Theorem 1 to derive the asymptotic normality for the optimal replacement period and for the optimal cost. Indeed, τ_0^* and C_0^* are the solutions of implicit equations and the classical delta-method cannot be anymore used. Thus, an analogous tool for this situation is required. It has been considered by Benichou and Gail [5].

Theorem 3. *Let $(X_n)_{n \in \mathbb{N}^*}$ be as in the Theorem 1. Let $\mu_X \in \mathbb{R}^p$ and $\mu_Y \in \mathbb{R}^q$. Let $g_1, \dots, g_q$ a set of q continuous functions from $\mathbb{R}^p \times \mathbb{R}^q$ into $\mathbb{R}$ with continuous first partial derivatives in an open set containing (μ_X, μ_Y) . Let Y_n be the $\mathbb{R}^q$ -valued random vectors satisfying $g_r(X_n, Y_n) = 0$ for all $r \in \{1, \dots, q\}$. Let $J_{x,y}$*

be the $q \times q$ matrix with elements $\frac{\partial g_i}{\partial y_j}(x, y)$ and let $H_{x,y}$ be the $q \times p$ matrix with elements $\frac{\partial g_i}{\partial x_j}(x, y)$. If $|J_{\mu_X, \mu_Y}| \neq 0$ and if each row of $J_{\mu_X, \mu_Y}^{-1} H_{\mu_X, \mu_Y}$ contain at least one nonzero element, then

$$\sqrt{n}(Y_n - \mu_Y) \xrightarrow[n \rightarrow \infty]{d} N(0, J_{\mu_X, \mu_Y}^{-1} H_{\mu_X, \mu_Y} \Sigma H_{\mu_X, \mu_Y}^T (J_{\mu_X, \mu_Y}^{-1})^T).$$

In fact, Theorem 2 holds for any replacement policy depending only the replacement period τ .

3.2. Periodic replacement policy

As an application of Theorem 3, one can prove that $\hat{\tau}_n^*$ is also asymptotically an estimator of τ_0^* under some regularity assumptions on the function ϕ .

Theorem 4. Assume that Equation (10) is satisfied and that $\theta \mapsto \phi(\tau; \theta)$ is differentiable for any $\tau > 0$. Let $\nabla_\theta \phi(\tau; \theta)$ be the gradient vector of ϕ (with respect to θ). If $\nabla_\theta \phi(\tau; \theta)$ is continuous with $\nabla_\theta \phi(\tau; \theta_0) \neq 0_{\mathbb{R}^p}$ and if $f_T(\tau_0^*; \theta_0) \neq 0$, then $\hat{\tau}_n^*$ is an asymptotic normal estimator of τ_0^* :

$$\sqrt{n}(\hat{\tau}_n^* - \tau_0^*) \xrightarrow[n \rightarrow \infty]{d} \mathcal{N}(0, \sigma_{opt.period}^2), \quad (11)$$

where $\sigma_{opt.period}^2$ is given by:

$$\sigma_{opt.period}^2 = \frac{\nabla_\theta \phi(\tau_0^*; \theta_0) \Sigma(\theta_0) \nabla_\theta \phi(\tau_0^*; \theta_0)^T}{[\tau_0^* f_T(\tau_0^*; \theta_0)]^2}. \quad (12)$$

Proof: First the partial derivative of ϕ with respect to τ is computed. From Equation (1), the following expression is derived:

$$\partial_\tau \phi(\tau; \theta) = \tau f_T(\tau; \theta),$$

where ∂_τ denotes the partial derivative with respect to τ . Using Theorem 3 and the expression of $\partial_\tau \phi$ the equations (11) and (12) are derived. $\diamond$

Similarly, under the same assumptions, the asymptotic normality of the optimal cost $\hat{C}_n^*$ can be also established.

Theorem 5. Assume that Equation (10) is satisfied and that $\theta \mapsto \psi(C^*; \theta)$ is differentiable for any $C^* > 0$. Let $\nabla_\theta \psi(C^*; \theta)$ be the gradient vector of ψ (with respect to θ). If $\nabla_\theta \psi(C^*; \theta)$ is continuous with $\nabla_\theta \psi(C_0^*; \theta_0) \neq 0_{\mathbb{R}^p}$, then $\hat{C}_n^*$ is an asymptotic normal estimator of C_0^* :

$$\sqrt{n}(\hat{C}_n^* - C_0^*) \xrightarrow[n \rightarrow \infty]{d} \mathcal{N}(0, \sigma_{opt.cost}^2),$$

where $\sigma_{opt.cost}^2$ is given by:

$$\sigma_{opt.cost}^2 = \left(\frac{c_u}{\tau_0^*} \right)^2 \nabla_\theta \phi(\tau_0^*; \theta_0) \Sigma(\theta_0) \nabla_\theta \phi(\tau_0^*; \theta_0)^T.$$

Proof: The results can be obtained from two ways. First, by applying the classical delta-method (see Theorem 1) to Equations (2),(10) and (11) and using the expression of $\sigma_{opt.period}^2$ given in Theorem 4. It leads to a first expression of the asymptotic variance $\sigma_{opt.cost}^2$ which is equal to:

$$\begin{aligned}\sigma_{opt.cost}^2 &= c_u^2 [f_T(\tau^*; \theta_0)]^2 \sigma_{opt.period}^2 \\ &= \left(\frac{c_u}{\tau_0^*} \right)^2 \nabla_\theta \phi(\tau_0^*; \theta_0) \Sigma(\theta_0) \nabla_\theta \phi(\tau_0^*; \theta_0)^T\end{aligned}\quad (13)$$

In the second way, by application of the delta-method for implicitly defined random variables (see Theorem 3), to Equations (10) and (3) the asymptotic variance can be written as a function of C^* . To this purpose, one shall compute the partial derivative of ψ given in Equation (3) with respect to C^* :

$$\partial_{C^*} \psi(C^*; \theta) = \frac{1}{c_u} F_T^{-1}(C^*/c_u; \theta) = \frac{\tau^*}{c_u}$$

It leads to the second expression of $\sigma_{opt.cost}^2$:

$$\sigma_{opt.cost}^2 = \left(\frac{c_u}{\tau_0^*} \right)^2 \nabla_\theta \psi(C_0^*; \theta_0) \Sigma(\theta_0) \nabla_\theta \psi(C_0^*; \theta_0)^T. \quad (14)$$

Of course, the two expressions (13) and (14) are equivalent. Indeed, ψ is derived from ϕ only by a change of variable with respect to τ , and not with respect to θ . Consequently, gradient vectors of ϕ and ψ with respect to θ are identical. Note that the gradient function (with respect to θ) involved in the above asymptotic variance can also be expressed as follows:

$$\nabla_\theta \psi(C_0^*; \theta_0) = \int_0^{C_0^*/c_u} \nabla_\theta F_T^{-1}(u; \theta_0) du.$$

◇

3.3. Age replacement policy

Similarly to the periodic replacement policy, under the conditions that the asymptotic normality holds for $\hat{\tau}_n^*$ using Theorem 3, the following theorem is proposed.

Theorem 6. Assume that Equation (10) is satisfied and that $\theta \mapsto \phi(\tau; \theta)$ is differentiable for any $\tau > 0$. Let $\nabla_\theta \phi(\tau; \theta)$ be the gradient vector of ϕ (with respect to θ). If $\nabla_\theta \phi(\tau; \theta)$ is continuous with $\nabla_\theta \phi(\tau; \theta_0) \neq 0_{\mathbb{R}^p}$ and if $\partial_\tau h_T(\tau_0^*; \theta_0) \neq 0$, then $\hat{\tau}_n^*$ is an asymptotic normal estimator of τ_0^* :

$$\sqrt{n} (\hat{\tau}_n^* - \tau_0^*) \xrightarrow[n \rightarrow \infty]{d} \mathcal{N}(0, \sigma_{opt.period}^2), \quad (15)$$

where $\sigma_{opt.period}^2$ is given by:

$$\sigma_{opt.period}^2 = \frac{\nabla_\theta \phi(\tau_0^*; \theta_0) \Sigma(\theta_0) \nabla_\theta \phi(\tau_0^*; \theta_0)^T}{\left[\partial_\tau h_T(\tau_0^*; \theta_0) \int_0^{\tau_0^*} S_T(u; \theta_0) du \right]^2}. \quad (16)$$

Proof: To apply Theorem 3 to $\hat{\tau}_n^*$, first one has to compute the partial derivative of ϕ with respect to τ :

$$\begin{aligned}\partial_\tau \phi(\tau; \theta) &= \partial_\tau h_T(\tau; \theta) \int_0^\tau S_T(u; \theta) du + h_T(\tau; \theta) S_T(\tau; \theta) - f_T(\tau; \theta) \\ &= \partial_\tau h_T(\tau; \theta) \int_0^\tau S_T(u; \theta) du\end{aligned}$$

where $\partial_\tau h$ is the derivative of h with respect to τ . The expressions (15) and (16) are then derived using Theorem 3 and the expression of $\partial_\tau \phi$. $\diamond$

To go through the asymptotic normality for the optimal cost $\hat{C}_n^*$ one proposes the following result.

Theorem 7. Assume that Equation (10) is satisfied and that $\theta \mapsto \psi(C^*; \theta)$ is differentiable for any $C^* > 0$. Let $\nabla_\theta \psi(C^*; \theta)$ be the gradient vector of ψ (with respect to θ). If $\nabla_\theta \phi(\tau; \theta)$ is continuous with $\nabla_\theta \phi(\tau; \theta_0) \neq 0_{\mathbb{R}^p}$ and if $\partial_\tau h_T(\tau_0^*; \theta_0) \neq 0$, then $\hat{C}_n^*$ is an asymptotic normal estimator of C_0^* :

$$\sqrt{n} \left(\hat{C}_n^* - C_0^* \right) \xrightarrow[n \rightarrow \infty]{d} \mathcal{N}(0, \sigma_{opt.cost}^2),$$

where $\sigma_{opt.cost}^2$ is given by:

$$\sigma_{opt.cost}^2 = \frac{[c_f - c_{nf}]^2 \nabla_\theta \phi(\tau_0^*; \theta_0) \Sigma(\theta_0) \nabla_\theta \phi(\tau_0^*; \theta_0)^T}{\left[\int_0^{\tau_0^*} S_T(u; \theta_0) du \right]^2}.$$

Proof: As for the periodic replacement policy, there is two equivalent ways to prove its asymptotic normality. However, one will only consider the first way since the second one is less straightforward. Under the same assumptions as in the previous theorem, using equation (4), one can express the asymptotic variance $\sigma_{opt.cost}^2$ in function of $\sigma_{opt.period}^2$ as follows:

$$\sigma_{opt.cost}^2 = [(c_f - c_{nf}) \partial_\tau h_T(\tau_0^*; \theta_0)]^2 \sigma_{opt.period}^2.$$

and the theorem is proved. $\diamond$

3.4. Block replacement policy

In the same way as other policies, under the conditions that the asymptotic normality holds for $\hat{\tau}_n^*$ using Theorem 3, the following theorem is proposed.

Theorem 8. Assume that Equation (10) is satisfied and that $\theta \mapsto \phi(\tau; \theta)$ is differentiable for any $\tau > 0$. Let $\nabla_\theta \phi(\tau; \theta)$ be the gradient vector of ϕ (with respect to θ). If $\nabla_\theta \phi(\tau; \theta)$ is continuous with $\nabla_\theta \phi(\tau; \theta_0) \neq 0_{\mathbb{R}^p}$ and if $\partial_\tau m(\tau_0^*; \theta_0) = \partial m(\tau; \theta) / \partial \tau \neq 0$, then $\hat{\tau}_n^*$ is an asymptotic normal estimator of τ_0^* :

$$\sqrt{n} (\hat{\tau}_n^* - \tau_0^*) \xrightarrow[n \rightarrow \infty]{d} \mathcal{N}(0, \sigma_{opt.period}^2), \quad (17)$$

where $\sigma_{opt.period}^2$ is given by:

$$\sigma_{opt.period}^2 = \frac{\nabla_{\theta}\phi(\tau_0^*; \theta_0)\Sigma(\theta_0)\nabla_{\theta}\phi(\tau_0^*; \theta_0)^T}{[\tau_0^*\partial_{\tau}m(\tau_0^*; \theta_0)]^2}. \quad (18)$$

Proof: Similarly to the other case, Theorem 3 is applied to $\hat{\tau}_n^*$ and since the partial derivative of ϕ with respect to τ is equal to $\tau\partial_{\tau}m(\tau; \theta)$, the expressions (17) and (18) are straightforward. $\diamond$

The following result go through the asymptotic normality for the optimal cost $\hat{C}_n^*$.

Theorem 9. Assume that Equation (10) is satisfied and that $\theta \mapsto \psi(C^*; \theta)$ is differentiable for any $C^* > 0$. Let $\nabla_{\theta}\psi(C^*; \theta)$ be the gradient vector of ψ (with respect to θ). If $\nabla_{\theta}\phi(\tau; \theta)$ is continuous with $\nabla_{\theta}\phi(\tau; \theta_0) \neq 0_{\mathbb{R}^p}$ and if $\partial_{\tau}m(\tau_0^*; \theta_0) \neq 0$, then $\hat{C}_n^*$ is an asymptotic normal estimator of C_0^* :

$$\sqrt{n} \left(\hat{C}_n^* - C_0^* \right) \xrightarrow[n \rightarrow \infty]{d} \mathcal{N}(0, \sigma_{opt.cost}^2),$$

where $\sigma_{opt.cost}^2$ is given by:

$$\sigma_{opt.cost}^2 = \frac{c_f \nabla_{\theta}\phi(\tau_0^*; \theta_0)\Sigma(\theta_0)\nabla_{\theta}\phi(\tau_0^*; \theta_0)^T}{\tau_0^{*2}}.$$

Proof: As for the two other replacement policies, there is two equivalent ways to prove its asymptotic normality. However, one will only consider the first way since the second one is less straightforward. Under the same assumptions as in the previous theorem, using equation (4), one can express the asymptotic variance $\sigma_{opt.cost}^2$ in function of $\sigma_{opt.period}^2$ as follows:

$$\sigma_{opt.cost}^2 = [c_f \partial_{\tau}m_T(\tau_0^*; \theta_0)]^2 \sigma_{opt.period}^2.$$

and the theorem is proved. $\diamond$

4. Example of application to some time-to-failure distributions

By considering particular time-to-failure distributions, one illustrates the proposed sensitivity analysis only for the age and periodic policy (a similar study can be reproduced for the block replacement policy).

4.1. Exponential distribution

Assume that the time-to-failure T is exponentially distributed with unknown parameter $\lambda_0 \in \mathbb{R}_+ = \Theta$:

$$\forall t \geq 0, \quad S_T(t) = e^{-\lambda_0 t}.$$

Assume that the available data is a n -sample $T_1, \dots, T_n$ of time-to-failure distributed as T . The maximum likelihood estimator is given by the following natural estimator:

$$\hat{\lambda}_n = \frac{n}{\sum_{i=1}^n T_i}.$$

It is well-known that this estimator is convergent and asymptotically normal:

$$\sqrt{n}(\hat{\lambda}_n - \lambda_0) \xrightarrow[n \rightarrow \infty]{d} \mathcal{N}(0, \lambda_0^2).$$

Periodic replacement policy. First, the expression of the function ϕ is provided as follows:

$$\phi(\tau; \lambda) = \frac{1}{\lambda} - \frac{c_r}{c_u} - \left(\frac{1}{\lambda} + \tau \right) e^{-\lambda\tau}.$$

It follows that τ^* satisfies the following identity:

$$\left(\frac{1}{\lambda} + \tau^* \right) e^{-\lambda\tau^*} = \frac{1}{\lambda} - \frac{c_r}{c_u}.$$

Notice that, if $\lambda\tau$ is 'small', then $\phi(\tau; \lambda) \sim \lambda\tau^2 - \frac{c_r}{c_u}$ and $\tau^* \sim \sqrt{\frac{c_u}{\lambda c_r}}$. To compute the asymptotic variance $\sigma_{opt.period}^2$, one shall compute the partial derivative of ϕ with respect to λ :

$$\partial_\lambda \phi(\tau; \lambda) = -\frac{1}{\lambda^2} + \left(\frac{1}{\lambda^2} + \frac{\tau}{\lambda} + \tau^2 \right) e^{-\lambda\tau}.$$

At points $(\tau; \lambda) = (\tau_0^*; \lambda_0)$, after some basic computations, it leads to the following expression for the standard deviation (and not the variance):

$$\sigma_{opt.period} = \tau_0^* + \frac{1}{\lambda_0} + \frac{1}{\lambda_0^2 \tau_0^*} \left(1 - e^{\lambda_0 \tau_0^*} \right).$$

Finally, when $\lambda_0 \tau_0^*$ tends to zero, $\sigma_{opt.period} \sim \tau_0^*$.

Next, the expression of $\sigma_{opt.cost}$ should be derived. First,

$$\psi(C; \lambda) = -\frac{1}{\lambda} \int_0^{C/c_u} \log(1-u) du - \frac{c_r}{c_u}.$$

Hence, C^* satisfies the following identity:

$$-\frac{1}{\lambda} \int_0^{C^*/c_u} \log(1-u) du = \frac{c_r}{c_u}.$$

Alternatively, $C^* = c_u(1 - \exp(-\lambda_0 \tau_0^*))$ and $C^* \sim c_u \lambda_0 \tau_0^*$ when $\lambda_0 \tau_0^*$ tends to zero. Finally,

$$\sigma_{opt.cost} = c_u \lambda_0 e^{-\lambda_0 \tau_0^*} \left(\tau_0^* + \frac{1}{\lambda_0} + \frac{1}{\lambda_0^2 \tau_0^*} \left(1 - e^{\lambda_0 \tau_0^*} \right) \right)$$

To illustrate the obtained results for the exponential distribution, some numerical computations are performed. The parameters has been set as follows: $c_u = 100$, c_r varies from 20 to 90 with a step equal to 0.1 and λ varies from 0.01 to 0.1 with a step equal to 0.001. The variations of optimal replacement period τ^* and the optimal cost C^* and their associated asymptotic variance $\sigma_{opt.period}$ and $\sigma_{opt.cost}$ are depicted in Figure ??.

As expected, when the failure rate is high, the replacements are more often (the optimal replacement period decreases as λ grows). Trivially, the optimal maintenance cost increases with the raise of the replacement cost. This increase is more noticeable for high failure rates which will of course induce not only more preventive replacements but also more unavailability periods. The uncertainty around the optimal value is acceptable and gives relatively manageable confidence intervals. As the failure rate grows, one earns on precision (small standard deviations). The same behaviour is in view for the optimal cost but the asymptotic standard deviations are larger. This latter can be explained by the expressions of $\sigma_{opt.period}$ and $\sigma_{opt.cost}$. Moreover, the asymptotic standard deviations are more sensitive to the failure rate variations than the cost variations.

Age replacement policy. Due to the memory-less property of the exponential distribution, this replacement policy is irrelevant and optimal replacements occur only at failure (i.e. $\tau_0^* = \infty$). For any fixed λ , the function $t \mapsto \phi(\tau; \lambda)$ is constant and equal to $-c_f/(c_f - c_{nf})$.

4.2. Weibull distribution

Consider a more general case by assuming that the time-to-failure T is Weibull distributed with parameter $\theta_0 = (\alpha_0, \beta_0) \in \Theta = \mathbb{R}_+^2$:

$$\forall t \geq 0, \quad S_T(t; \alpha_0, \beta_0) = \exp \left(- \left(\frac{t}{\alpha_0} \right)^{\beta_0} \right) \quad \text{and}$$

$$f_T(t; \alpha_0, \beta_0) = \frac{\beta_0}{\alpha_0} \left(\frac{t}{\alpha_0} \right)^{\beta_0-1} \exp \left(- \left(\frac{t}{\alpha_0} \right)^{\beta_0} \right).$$

As for the previous example, the available data is a n -sample $T_1, \dots, T_n$ of time-to-failure distributed as T . The maximum likelihood estimators $\hat{\alpha}_n$ and $\hat{\beta}_n$ satisfy the following equations (see equations (27.71) and (21.72), page 656, in [18]):

$$\hat{\alpha}_n = \left(\frac{1}{n} \sum_{i=1}^n T_i^{\hat{\beta}_n} \right)^{1/\hat{\beta}_n}$$

and

$$\hat{\beta}_n = \left[\left(\sum_{i=1}^n T_i^{\hat{\beta}_n} \log T_i \right) \left(\sum_{i=1}^n T_i^{\hat{\beta}_n} \right)^{-1} - \frac{1}{n} \sum_{i=1}^n \log T_i \right]^{-1}.$$

The estimate vector is asymptotically normal with asymptotic variance given by

$$\Sigma(\theta_0) = \frac{1}{\alpha_0^2(\Psi_1(1) + \Psi(2)^2 - (1 + \Psi(1))^2)} \begin{pmatrix} (\alpha_0\beta_0)^2 & -\alpha_0(1 + \Psi(1)) \\ -\alpha_0(1 + \Psi(1)) & \frac{\Psi_1(1) + \Psi(2)^2}{\beta_0^2} \end{pmatrix}$$

where Ψ and Ψ_1 are respectively the digamma and the trigamma functions (see [15], for instance), $\Psi(1) = -\gamma$ (Euler's constant), $\Psi(2) = 1 + \Psi(1)$ and $\Psi_1(1) = \pi^2/6$.

Below, the integral of the survival function over a finite interval will be involved in the computations. Hence its expression is given here. From Equation (3.381.8) in [15],

$$\int_0^t S(u; \theta) du = \int_0^t \exp\left(-\left(\frac{u}{\alpha_0}\right)^{\beta_0}\right) du = \frac{\alpha_0}{\beta_0} \gamma\left(\beta_0^{-1}, \left(\frac{t}{\alpha_0}\right)^{\beta_0}\right) \quad (19)$$

where $\gamma(\cdot, \cdot)$ is the lower incomplete gamma function (see [1] or [15], for instance).

At last, for sake of simplicity, let us denote $\lambda_0 = \frac{\tau}{\alpha_0}$.

Periodic replacement policy.. For this policy and for the Weibull distribution, the cost function is:

$$\begin{aligned} C(\tau; \alpha_0, \beta_0) &= \frac{1}{\tau} \left\{ c_r + c_u \int_0^\tau \left[1 - \exp\left(-\left(\frac{u}{\alpha_0}\right)^{\beta_0}\right) \right] du \right\} \\ &= \frac{1}{\tau} \left\{ c_r + c_u \tau - c_u \frac{\alpha_0}{\beta_0} \gamma\left(\beta_0^{-1}, \lambda_0^{\beta_0}\right) \right\}. \end{aligned}$$

From the above expression of the cost function, its partial derivatives with respect to α and β are:

$$\partial_\alpha C(\tau; \alpha_0, \beta_0) = c_u \left\{ -\frac{1}{\tau\beta_0} \gamma\left(\beta_0^{-1}, \lambda_0^{\beta_0}\right) + \frac{1}{\alpha_0} \exp\left(-\lambda_0^{\beta_0}\right) \right\}$$

and

$$\begin{aligned} \partial_\beta C(\tau; \lambda, \beta_0) &= \frac{c_u}{\lambda\beta_0^3} \left\{ \beta_0 \gamma(\beta_0^{-1}, \lambda^{\beta_0}) - \beta_0^2 \lambda \ln(\lambda) e^{-\lambda^{\beta_0}} - \beta_0 \Gamma(\beta_0^{-1}, \lambda^{\beta_0}) \ln(\lambda) \right. \\ &\quad \left. - G_{2,3}^{3,0} \left(\lambda^{\beta_0} \middle| \begin{matrix} 1 & 1 \\ 0 & 0 \end{matrix} \begin{matrix} 1 \\ \beta_0^{-1} \end{matrix} \right) + \Psi(\beta_0^{-1}) \Gamma(\beta_0^{-1}) \right\} \end{aligned}$$

where G is the Meijer's G function [15]. This function can be expressed in function of the hypergeometric function ${}_2F_2$ and gamma related functions as follows (see, e.g., [8, formula 42, page 645]):

$$G_{2,3}^{3,0} \left(z \middle| \begin{matrix} 1 & 1 \\ 0 & 0 \end{matrix} \begin{matrix} 1 \\ a \end{matrix} \right) = \Gamma(a) [\psi(a) - \log(a)] + \left(\frac{z}{a}\right)^2 {}_2F_2(a, a; a+1, a+1; -z).$$

Now, let us consider the function ϕ . Using the identity given in Equation (19):

$$\phi(\tau; \alpha_0, \beta_0) = -\tau \exp(-\lambda_0^{\beta_0}) + \frac{\alpha_0}{\beta_0} \gamma(\beta_0^{-1}, \lambda_0^{\beta_0}) - \frac{c_r}{c_u}.$$

Therefore, the first order partial derivatives of ϕ are as follows:

$$\partial_{\alpha_0} \phi(\tau; \alpha_0, \beta_0) = -\lambda_0 \left(1 + \beta_0 \lambda_0^{\beta_0}\right) \exp(-\lambda_0^{\beta_0}) + \beta_0^{-1} \gamma(\beta_0^{-1}, \lambda_0^{\beta_0}) \quad (20)$$

and

$$\begin{aligned} \partial_{\beta_0} \phi(\tau; \alpha_0, \beta_0) &= \frac{\tau_0}{\beta_0} \log(\lambda_0) \exp(-\lambda_0^{\beta_0}) \left(1 + \beta_0 \lambda_0^{\beta_0}\right) + \\ &\frac{\alpha_0}{\beta_0^3} \left(-\beta_0 \gamma(\beta_0^{-1}, \lambda_0^{\beta_0}) + \beta_0 \log(\lambda_0) \Gamma(\beta_0^{-1}, \lambda_0^{\beta_0}) + G_{2,3}^{3,0} \left(\lambda_0^{\beta_0} \begin{vmatrix} 1 & 1 \\ 0 & 0 \end{vmatrix} \beta_0^{-1} \right) - \Psi(\beta_0^{-1}) \Gamma(\beta_0^{-1}) \right) \end{aligned} \quad (21)$$

Moreover, from Equation (19),

$$\tau_0^* = \alpha_0 (-\log(1 - C^*/c_u))^{1/\beta_0}$$

where C^* is the solution of

$$\alpha_0 \int_0^{C^*/c_u} (-\log(1-v))^{\beta_0^{-1}} dv - \frac{c_r}{c_u} = 0 \Leftrightarrow \gamma\left(\beta_0^{-1} - 1, -\log\left(1 - \frac{C^*}{c_u}\right)\right) = \frac{c_r}{\alpha_0 c_u}$$

Let be $\nu = -\log(1 - \frac{C^*}{c_u})$. By replacing $\tau_0^* = \alpha_0 \nu^{\beta_0^{-1}}$ in the partial derivative of ϕ with respect to α and to β , see Equations (20) and (21), the following expressions are obtained:

$$\partial_{\alpha} \phi(\alpha_0 \nu^{\beta_0^{-1}}; \alpha_0, \beta_0) = \beta_0^{-1} \gamma(\beta_0^{-1}, \nu) - (\beta_0 \nu + 1) \left(1 - \frac{C^*}{c_u}\right) \nu^{\beta_0^{-1}} \quad (22)$$

and

$$\begin{aligned} \partial_{\beta} \phi(\alpha_0 \nu^{\beta_0^{-1}}; \alpha_0, \beta_0) &= \frac{\alpha_0}{\beta_0} \nu^{\beta_0^{-1}} \left(1 - \frac{C^*}{c_u}\right) \left(\nu + \frac{1}{\beta_0}\right) \log(\nu) \\ &+ \frac{\alpha_0}{\beta_0^3} \left(-\beta_0 \gamma(\beta_0^{-1}, \nu) + \Gamma(\beta_0^{-1}, \nu) \log(\nu) + G_{2,3}^{3,0} \left(\nu \begin{vmatrix} 1 & 1 \\ 0 & 0 \end{vmatrix} \beta_0^{-1} \right) - \Psi(\beta_0^{-1}) \Gamma(\beta_0^{-1}) \right) \end{aligned} \quad (23)$$

Therefore, by Theorems 4 and 5, one can derive the asymptotic variance of $\hat{\tau}_n^*$ by replacing Equations (22) and (23) as follows:

$$\sigma_{opt.period}^2 = \frac{\nabla_{\theta} \phi(\tau_0^*; \theta_0) \Sigma(\theta_0) \nabla_{\theta} \phi(\tau_0^*; \theta_0)^T}{\left[\beta_0 \left(1 - \frac{C^*}{c_u}\right) \log\left(1 - \frac{C^*}{c_u}\right) \right]^2}.$$

Similarly, the asymptotic variance of $\hat{C}_n^*$ is obtained as follows:

$$\sigma_{opt.cost}^2 = \frac{c_u^2 \nabla_{\theta} \phi(\tau_0^*; \theta_0) \Sigma(\theta_0) \nabla_{\theta} \phi(\tau_0^*; \theta_0)^T}{\left[\alpha_0 \left(-\log \left(1 - \frac{C^*}{c_u} \right) \right)^{1/\beta_0} \right]^2}.$$

To end with this case, some numerical computations are performed. The parameters has been set as follows: $c_u = 100$, c_r varies from 20 to 90 with a step equal to 0.1.

The optimal replacement period and its asymptotic standard deviation in function of preventive replacement cost c_r and the shape parameter β for three different values of $\alpha \in \{1, 5, 10\}$ are plotted on Figure ???. The optimal replacement period and its standard deviation in function of c_r and β when $\alpha = 1$ are depicted in Figure ??. The following points can be highlighted.

- Replacements are less frequent when the replacements are expensive.
- The optimal replacement period is an increasing function of the scale parameter α . As the mean time to failure grows with α the replacements become less frequent. For small values of α , the optimal replacement period is not sensitive to variations of β .
- The standard deviation is not very sensitive to the replacement cost unit.
- The shape parameter β impacts significantly the standard deviation. The standard deviation of the optimal replacement period is an increasing function of β for $\beta > 2$ and a non-increasing function of β for $\beta \in [1, 2]$.
- Moreover, standard deviation is not very sensitive to variations of α .

In sum, the confidence bounds around the optimal replacement period are not very large and depend essentially on the shape parameter β . Decision making based on estimated parameters rather than true unknown lifetime distribution parameters has not a significant impact on the replacement procedure.

The optimal maintenance cost $\hat{C}_n^*$ and its standard deviation are depicted on Figures ?? as functions of the replacement cost c_r and shape parameter β for three different values of $\alpha \in \{1, 5, 10\}$.

- As expected the optimal cost is an increasing function of the replacement cost unit. Indeed, as the replacement cost increases all types of replacement, preventive or corrective, are costly and the average total cost increases.
- When α increases, the mean time to failure increases and the replacements are less often which decreases the optimal average maintenance cost. For small values of α , the optimal cost is not sensitive to variations of β .
- As in the replacement period case, the standard deviation is not very sensitive to the maintenance cost unit. It decreases as α grows and it is

an increasing function of β . In other words, when the mean time to failure increases, the confidence bound around the true optimal cost (based on true unknown parameters) get smaller.

It is important to highlight, the large standard deviation associated to the optimal cost. It means that even with a very large data sample, the cost difference between the maintenance policy based on parameter estimates and the policy based on the true unknown parameters is significant.

Age replacement policy. From Equation (19), it follows that

$$\phi(t; \alpha_0, \beta_0) = \left(\frac{t}{\alpha_0}\right)^{\beta_0-1} \gamma\left(\beta_0^{-1}, \left(\frac{t}{\alpha_0}\right)^{\beta_0}\right) + \exp\left(-\left(\frac{t}{\alpha_0}\right)^{\beta_0}\right) - \frac{c_f}{c_f - c_{nf}}.$$

Hence, the (unknown) optimal replacement period τ_0^* satisfies the following equation:

$$\phi(\tau_0^*; \alpha_0, \beta_0) = \lambda_0^{\beta_0-1} \gamma\left(\beta_0^{-1}, \lambda_0^{\beta_0}\right) + \exp\left(-\lambda_0^{\beta_0}\right) - \frac{c_f}{c_f - c_{nf}} = 0.$$

Using Equation (5), since $h_T^{-1}(u; \alpha_0, \beta_0) = \alpha_0(\alpha_0 u / \beta_0)^{\frac{1}{\beta_0-1}}$, the optimal replacement period can be as well obtained as follows:

$$\tau_0^* = \alpha_0 \left(\frac{\alpha_0 C_0^*}{\beta_0(c_f - c_{nf})} \right)^{\frac{1}{\beta_0-1}}.$$

Moreover, C_0^* is the solution of the following equation:

$$\frac{\alpha_0}{\beta_0} C_0^* \gamma\left(\beta_0^{-1}, \left(\frac{\alpha_0 C_0^*}{\beta_0(c_f - c_{nf})}\right)^{\frac{\beta_0}{\beta_0-1}}\right) + (c_n - c_{nf}) \exp\left(-\left(\frac{\alpha_0 C_0^*}{\beta_0(c_f - c_{nf})}\right)^{\frac{\beta_0}{\beta_0-1}}\right) - c_f = 0. \quad (24)$$

In order to obtain the the asymptotic variance of the optimal replacement period and optimal cost, the derivatives of the function ϕ evaluated at point τ_0^* are expressed as follows:

$$\partial_\alpha \phi(\tau_0^*; \alpha_0, \beta_0) = -\frac{(\beta_0 - 1)\lambda_0^{\beta_0}}{\tau_0^*} \gamma\left(\beta_0^{-1}, \lambda_0^{\beta_0}\right)$$

and

$$\begin{aligned} \partial_\beta \phi(\tau_0^*; \alpha_0, \beta_0) &= \lambda_0 \ln \lambda_0 \exp\left(-\lambda_0^{\beta_0}\right) (1 - \lambda_0^{\beta_0-1}) + \\ &\quad \beta_0^{-2} \lambda_0^{\beta_0-1} \left[\beta_0^2 \gamma\left(\beta_0^{-1}, \lambda_0^{\beta_0}\right) \log \lambda_0 + \beta_0 \Gamma\left(\beta_0^{-1}, \lambda_0^{\beta_0}\right) \log \lambda_0 \right. \\ &\quad \left. + G_{2,3}^{3,0}\left(\lambda_0^{\beta_0} \middle| \begin{matrix} 1 & 1 \\ 0 & 0 \end{matrix} \beta_0^{-1} \right) - \Psi(\beta_0^{-1}) \Gamma(\beta_0^{-1}) \right] \end{aligned}$$

Since

$$\partial_\tau h_T(\tau_0^*; \theta_0) \int_0^{\tau_0^*} S_T(u; \theta_0) du = \frac{(\beta_0 - 1)\lambda_0^{\beta_0 - 2}}{\alpha_0} \gamma(\beta_0^{-1}, \lambda_0^{\beta_0}),$$

the asymptotic variance $\sigma_{opt.period}^2$ can be deduced from Theorem 6 and consequently $\sigma_{opt.cost}^2$ is given by:

$$\sigma_{opt.cost}^2 = \left[(c_f - c_{nf}) \frac{\beta_0(\beta_0 - 1)\lambda_0^{\beta_0 - 2}}{\alpha_0^2} \right]^2 \sigma_{opt.period}^2.$$

Impact of the scale parameter Now one goes through scaling properties of the various quantities of interest with respect to the scale parameter α_0 . To this end, one shall denote these quantities by $\tau_0^*(\alpha_0)$, $C_0^*(\alpha_0)$, $\sigma_{opt.period}^2(\alpha_0)$ and $\sigma_{opt.cost}^2(\alpha_0)$. Notice that the function ϕ depends on t and α_0 only through their ratio t/α_0 . So, it follows that $\tau_0^*(\alpha_0) = \alpha_0 \tau_0^*(1)$. As a consequence, $\lambda_0 = \tau_0^*(\alpha_0)/\alpha_0 = \tau_0^*(1)$ does not depend on α_0 . Similarly, Equation (24) depends on $C_0^*(\alpha_0)$ and α_0 only through their product $\alpha_0 C_0^*(\alpha_0)$ and so $C_0^*(\alpha_0) = C_0^*(1)/\alpha_0$. Now let us consider the variance. From the previous expressions,

$$\partial_\alpha \phi(\tau_0^*(\alpha_0); \alpha_0, \beta_0) = \frac{1}{\alpha_0} \partial_\alpha \phi(\tau_0^*(1); 1, \beta_0)$$

and

$$\partial_\beta \phi(\tau_0^*(\alpha_0); \alpha_0, \beta_0) = \partial_\beta \phi(\tau_0^*(1); 1, \beta_0)$$

since, for the latter, it depends on λ_0 and not in α_0 . Consequently, using the expression of $\Sigma(\alpha_0, \beta_0)$ given above and from the above expressions,

$$\begin{aligned} \nabla_\theta \phi(\tau_0^*(\alpha_0, \beta_0); \alpha_0, \beta_0) \Sigma(\alpha_0, \beta_0) \nabla_\theta \phi(\tau_0^*(\alpha_0, \beta_0); \alpha_0, \beta_0)^T \\ = \frac{1}{\alpha_0^2} \nabla_\theta \phi(\tau_0^*(1, \beta_0); 1, \beta_0) \Sigma(1, \beta_0) \nabla_\theta \phi(\tau_0^*(1, \beta_0); 1, \beta_0)^T \end{aligned}$$

Moreover, from the expression of the hazard function for the Weibull distribution,

$$\partial_\tau h_T(\tau_0^*(\alpha_0, \beta_0); \alpha_0, \beta_0) = \frac{1}{\alpha_0^2} \partial_\tau h_T(\tau_0^*(1, \beta_0); 1, \beta_0).$$

At last, it is straightforward to get that, for the Weibull distribution,

$$\int_0^{\tau_0^*(\alpha_0, \beta_0)} S_T(u; \alpha_0, \beta_0) du = \alpha_0 \int_0^{\tau_0^*(1, \beta_0)} S_T(u; 1, \beta_0) du.$$

Consequently, jointly these identities leads to the following ones:

$$\sigma_{opt.period}^2(\alpha_0) = \sigma_{opt.period}^2(1)$$

and

$$\sigma_{opt.cost}^2(\alpha_0) = \frac{1}{\alpha_0^4} \sigma_{opt.cost}^2(1).$$

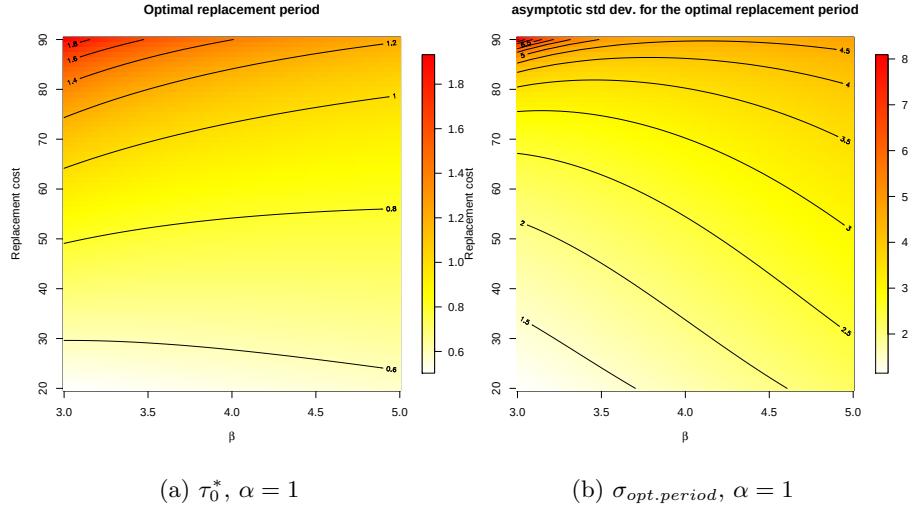


Figure 1: Age replacement policy: case of Weibull distribution, the optimal replacement period and its asymptotic standard deviation for different values of $\beta \in [3, 5]$ and $\alpha = 1$

To end with this case, some numerical computations are carried out. As for the previous replacement policy, the parameters has been set as follows: $c_u = 100$, c_r varies from 20 to 90 with a step equal to 0.1. Since the standard deviations are proportional to α , only the numerical results considering $\alpha = 1$ are depicted. The optimal replacement period and its standard deviation, τ^* and $\sigma_{opt.period}$, are plotted on Figure 1 in function of c_r and β for $\alpha = 1$. In Figure 2, the optimal maintenance cost and its standard deviation are depicted. The following remarks can be pointed out.

- The replacements are more frequent as the mean time to failure decreases (β increases).
- The replacement are less frequent when they are costly.
- The standard deviation of the optimal replacement period increases with the replacement cost and the mean time to failure. In other words, as β increases the uncertainty around the optimal replacement period decreases.
- The optimal maintenance cost is not sensitive the the shape parameter β and it increases with the replacement cost.
- The uncertainty around optimal cost increases with the replacement cost and β .
- The standard deviation of the optimal cost takes large values.

In sum, even with a very large data sample, the cost and replacement period difference between the maintenance policy based on parameter estimates and the policy based on the true unknown parameters are significant.

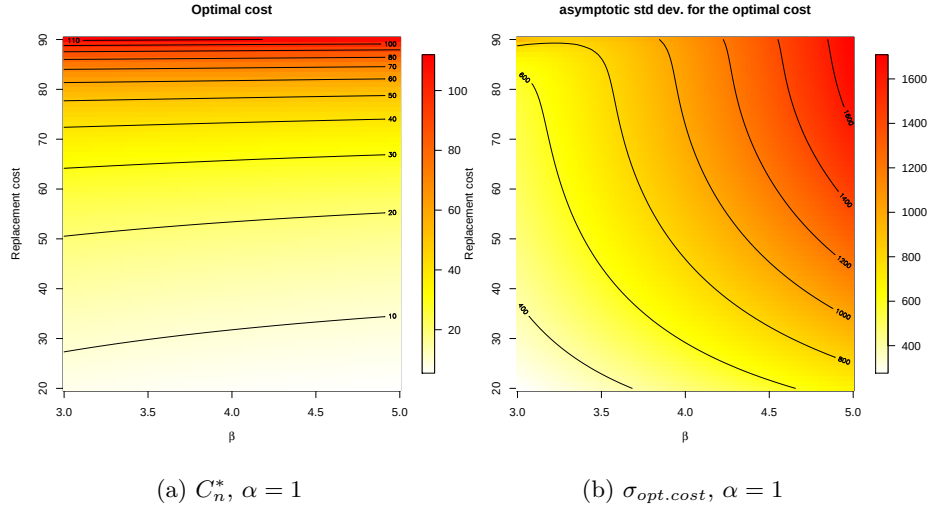


Figure 2: Age replacement policy: case of Weibull distribution, the optimal average total cost and its asymptotic standard deviation with respect to different values of $\beta \in [3, 5]$ and $\alpha = 1$

5. Conclusions and discussions

This paper proposed a sensitivity analysis of two replacement policies, namely the periodic replacement policy and the age replacement policy. The aim of the paper has been to quantify how uncertainty on parameters of the time-to-failure distribution (which are unknown and have to be estimated) has an impact on optimal quantities of interest for a given policy. In particular, it is shown that, provided that the estimator of the parameters of the time-to-failure distribution are asymptotically normal, the estimator of the optimal replacement period and the one of the optimal cost are also asymptotically normal. The results are illustrated by focusing on two different time-to-failure distributions, the exponential and the Weibull distributions. These results permit in practice to have confidence bound for the bias of the optimal maintenance cost related to the true but unknown lifetime distribution parameters and the optimal cost based on the lifetime parameter estimators. Based on these bounds decision makers can evaluate the excess cost when large data is available and true lifetime parameters are unknown and use this information to support their decision.

Since our results are based on asymptotic properties, it may be quite difficult to use our methodology in practice. Indeed, in reliability studies, time-to-failure are frequently right-censored and/or the sample size are rather small. In the first case, our approach can be adapted without any difficulty. In fact, one has just to establish the asymptotic behaviours of the estimator. For instance, consider the type I right-censoring scheme with the exponential distribution (with parameter λ) for the time-to-failure distribution and the same censoring instant c for all units. In such case, it can be easily proved that the maximum

likelihood estimator $\hat{\lambda}_n$ is asymptotically normally distributed with zero mean and variance $\lambda^2/(1 - e^{-\lambda c})$. Consequently, for the periodic replacement policy, $\sigma_{opt.period}$ is now equal to

$$\sigma_{opt.period} = \left[\tau_0^* + \frac{1}{\lambda_0} + \frac{1}{\lambda_0^2 \tau_0^*} \left(1 - e^{\lambda_0 \tau_0^*} \right) \right] / \sqrt{1 - e^{-\lambda_0 c}}.$$

Of course, when c tends to infinity (no right-censoring occurs), the above expression tends to the one obtained earlier in this paper. Whatever censorship or not, when the sample size is not enough large, such results cannot be no more applied. An alternative solution consists in using bootstrap method.

The approach developed here could also be used for a sensitivity analysis of a different replacement policy or of more complex policies like condition-based maintenance policy (see [6], for instance). In these frameworks further analytical expressions should be provided.

Acknowledgments

This work was supported by the French National Agency of Research (ANR), project AMMSI, number ANR-2011-BS01-021.

References

- [1] M. Abramowitz and I. Stegun. *Handbook of mathematical functions with formulas, graphs, and mathematical tables*. National Bureau of Standards, 1972.
- [2] R.D. Baker and P.A. Scarf. Can models fitted to small data samples lead to maintenance policies with near-optimum cost? *IMA Journal of Management Mathematics*, 6(1): 3-12, 1995.
- [3] T. Bedford and C. Bunea. The effect of model uncertainty on maintenance optimization. *IEEE Transactions on Reliability*, 51(4): 486–493, 2002.
- [4] T. Bedford and K.J. Wilson and A. Daneshkhah. Assessing parameter uncertainty on coupled models using minimum information methods. *Reliability Engineering and System Safety*, 125: 3–12, 2014.
- [5] J. Benichou and M.H. Gail. A delta method for implicitly defined random variables. *The American Statistician*, 43(1): 41–44, 1989.
- [6] C. Bérenguer C. and A. Grall and L. Dieulle and M. Roussignol. Maintenance policy for a continuously monitored deteriorating system. *Probability in the Engineering and Informational Sciences*, 17(2): 235–250, 2003.
- [7] J. Bernardo and A. Smith. *Bayesian theory*. Wiley, 1993.
- [8] Y.A. Brychkov. *Handbook of special functions: derivatives, integrals, series and other formulas*. Chapman and Hall/CRC Press, 2008.

- [9] G. Castella and R.L. Berger. *Statistical Inference*. Duxbury Advanced Series, Thomsom learning, 2002.
- [10] R. Barlow and L. Hunter. Optimum preventive maintenance policies. *Operation Research*, 8(1): 90–100, 1960.
- [11] B. de Jonge and W. Klingenberg and R. Teunter and T.Tinga. Optimum maintenance strategy under uncertainty in the lifetime distribution. *Reliability Engineering and System Safety*, 133: 59–67, 2015.
- [12] B. de Jonge and A.J Dijkstra and W. Romeijnders. Cost benefit of postponing time-based maintenance under distribution uncertainty. *Reliability Engineering and System Safety*, 140: 15–21, 2015.
- [13] W. Feller. *An Introduction to probability theory and its applications. Volume 2*. John Wiley and sons, 1971.
- [14] G. Glasser. The age replacement problem. *Technometrics*, 9(1): 83–91, 1967.
- [15] I. Gradshteyn and I. Ryzhik. *Table of integrals, series and products*. Academic Press, 1965.
- [16] M. Hamada and A. Wilson and C. Reese and H. Martz. *Bayesian reliability*. Springer, 2010.
- [17] X. Jiang and V. Makis and A. Jardine. Optimal repair/replacement policy for a general repair model. *Advances on Applied Probability*, 33(1): 206–222, 2001.
- [18] N.L Johnson and S. Kotz and N. Balakrishnan. *Continuous univariate distributions. Volume 1*. John Wiley and sons, 1994.
- [19] M. Kijima and H. Morimura and Y. Suzuki. Periodical replacement problem without assuming minimal repair. *European Journal of Operation Research*, 37(2): 194–203, 1988.
- [20] T. Mazzuchi and R. Soyer. A Bayesian perspective on some replacement strategies. *Reliability Engineering System Safety*, 51(3): 295–303, 1996.
- [21] T. Nakagawa. *Maintenance theory of reliability*. Springer-Verlag, 2005.
- [22] T. Nakagawa and K. Yasui. Calculation of age-replacement with Weibull failure times. *IEEE Transactions on Reliability*, 30(2): 163–164, 1981.
- [23] A. O’Hagan. *Kendall’s advanced theory of statistics: Bayesian inference*. Edward Arnold, 1994.
- [24] A. Papadopoulos and C. Tsokos. Bayesian confidence bounds for the Weibull failure model. *IEEE Transactions on Reliability*, 24(1): 21–26, 1975.

- [25] C.R. Rao. *Linear Statistical Inference and its Applications*. John Wiley and sons, 1973.
- [26] E. Silver and C.N. Fiechter. A simple case of preventive maintenance decision-making with limited historical data. *International Journal of Production Economics*, 27(3): 241–250, 1992.
- [27] E. Silver and C.N. Fiechter. Preventive maintenance with limited historical data. *European Journal of Operational Research*, 82(1): 125-144, 1995.
- [28] H. Wang. A survey of maintenance policies of deteriorating systems. *European Journal of Operational Research*, 139: 469–489, 2002.